

PROJECT ZEPHYRAQUUS

ENGINEERING NOTEBOOK

PROTOTYPE V0.2 – ELECTRONICS LAYOUT

PROTOTYPE ASSEMBLY

Objective-002:

Finalize placement of all electrical components, establish wiring architecture, verify accessibility, and prepare the prototype for permanent assembly.

DG-002:

- Determine final placement of both SpeedyBee stacks.
- Route all motor wires with minimal excess.
- Position the battery to maintain the center of gravity.
- Verify that all components are accessible for maintenance.
- Plan the electrical architecture before soldering.

DD-002:

1. All eight propulsion motors were mounted to the prototype board using the shortest compatible mounting screws.
2. Motor orientation was selected such that the motor leads faced inward toward the planned ESC locations, minimizing wire length and simplifying future routing; shown in Fig. 2.
3. Motor mounting patterns were pressed into the cardboard to accurately transfer the mounting-hole locations prior to drilling.

ECE-002:

- Pressed the motor mounting points into the cardboard to create an indentation pattern for accurate hole placement.
- Used shorter screws for the mounting holes closer to the motor wires because of limited clearance. These locations were marked on the underside of the board to differentiate them from the standard screw positions.

Figure 2. Top and bottom views of the completed motor installation showing inward-facing motor leads, mounting-hole locations, and underside screw configuration prior to ESC installation.

